

3.2 TRANSISTOR CONSTRUCTION

The transistor is a three-layer semiconductor device consisting of either two *n*- and one *p*-type layers of material or two *p*- and one *n*-type layers of material. The former is called an *npn transistor*, while the latter is called a *pnp transistor*. Both are shown in Fig. 3.2 with the proper dc biasing. We will find in Chapter 4 that the dc biasing is necessary to establish the proper region of operation for ac amplification. The emitter layer is heavily doped, the base lightly doped, and the collector only lightly doped. The outer layers have widths much greater than the sandwiched *p*- or *n*-type material. For the transistors shown in Fig. 3.2 the ratio of the total width to that of the center layer is $0.150/0.001 = 150:1$. The doping of the sandwiched layer is also considerably less than that of the outer layers (typically, 10:1 or less). This lower doping level decreases the conductivity (increases the resistance) of this material by limiting the number of “free” carriers.

For the biasing shown in Fig. 3.2 the terminals have been indicated by the capital letters *E* for *emitter*, *C* for *collector*, and *B* for *base*. An appreciation for this choice of notation will develop when we discuss the basic operation of the transistor. The abbreviation BJT, from *bipolar junction transistor*, is often applied to this three-terminal device. The term *bipolar* reflects the fact that holes and electrons participate in the injection process into the oppositely polarized material. If only one carrier is employed (electron or hole), it is considered a *unipolar* device. The Schottky diode of Chapter 20 is such a device.

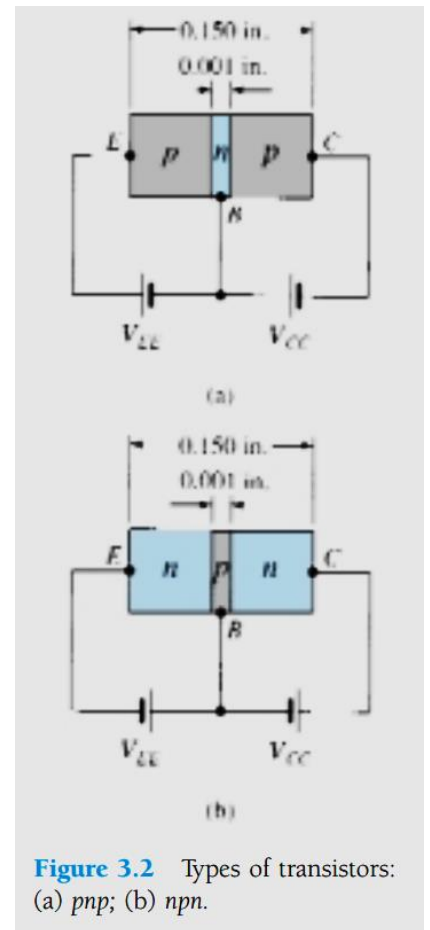


Figure 3.2 Types of transistors: (a) *pnp*; (b) *npn*.

3.3 TRANSISTOR OPERATION

The basic operation of the transistor will now be described using the *pnp* transistor of Fig. 3.2a. The operation of the *npn* transistor is exactly the same if the roles played by the electron and hole are interchanged. In Fig. 3.3 the *pnp* transistor has been redrawn without the base-to-collector bias. Note the similarities between this situation and that of the *forward-biased* diode in Chapter 1. The depletion region has been reduced in width due to the applied bias, resulting in a heavy flow of majority carriers from the *p*- to the *n*-type material.

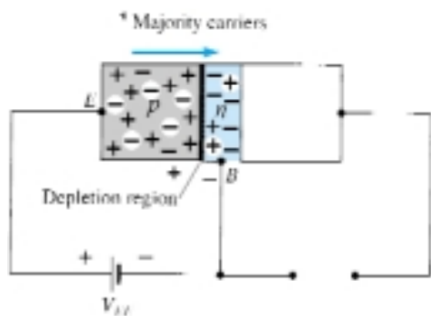


Figure 3.3 Forward-biased junction of a *pnp* transistor.

Let us now remove the base-to-emitter bias of the *pnp* transistor of Fig. 3.2a as shown in Fig. 3.4. Consider the similarities between this situation and that of the *reverse-biased* diode of Section 1.6. Recall that the flow of majority carriers is zero, resulting in only a minority-carrier flow, as indicated in Fig. 3.4. In summary, therefore:

One p-n junction of a transistor is reverse biased, while the other is forward biased.

In Fig. 3.5 both biasing potentials have been applied to a *pnp* transistor, with the resulting majority- and minority-carrier flow indicated. Note in Fig. 3.5 the widths of the depletion regions, indicating clearly which junction is forward-biased and which is reverse-biased. As indicated in Fig. 3.5, a large number of majority carriers will diffuse across the forward-biased *p-n* junction into the *n*-type material. The question then is whether these carriers will contribute directly to the base current I_B or pass directly into the *p*-type material. Since the sandwiched *n*-type material is very thin and has a low conductivity, a very small number of these carriers will take this path of high resistance to the base terminal. The magnitude of the base current is typically on the order of microamperes as compared to milliamperes for the emitter and collector currents. The larger number of these majority carriers will diffuse across the reverse-biased junction into the *p*-type material connected to the collector terminal as indicated in Fig. 3.5. The reason for the relative ease with which the majority carriers can cross the reverse-biased junction is easily understood if we consider that for the reverse-biased diode the injected majority carriers will appear as minority carriers in the *n*-type material. In other words, there has been an *injection* of minority carriers into the *n*-type base region material. Combining this with the fact that all the minority carriers in the depletion region will cross the reverse-biased junction of a diode accounts for the flow indicated in Fig. 3.5.

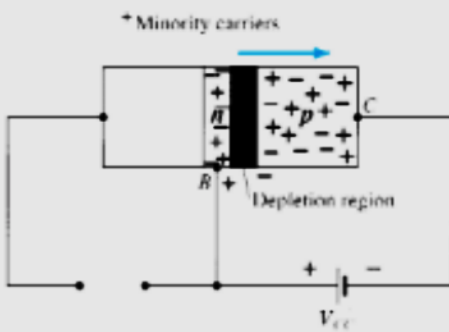


Figure 3.4 Reverse-biased junction of a *pnp* transistor.

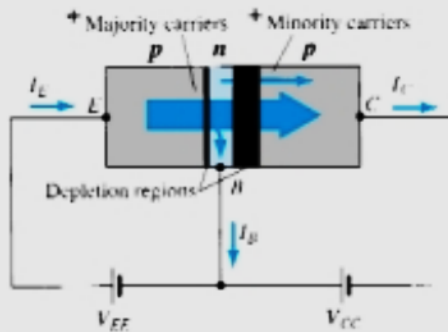


Figure 3.5 Majority and minority carrier flow of a *pnp* transistor.

Applying Kirchhoff's current law to the transistor of Fig. 3.5 as if it were a single node, we obtain

$$I_E = I_C + I_B \quad (3.1)$$

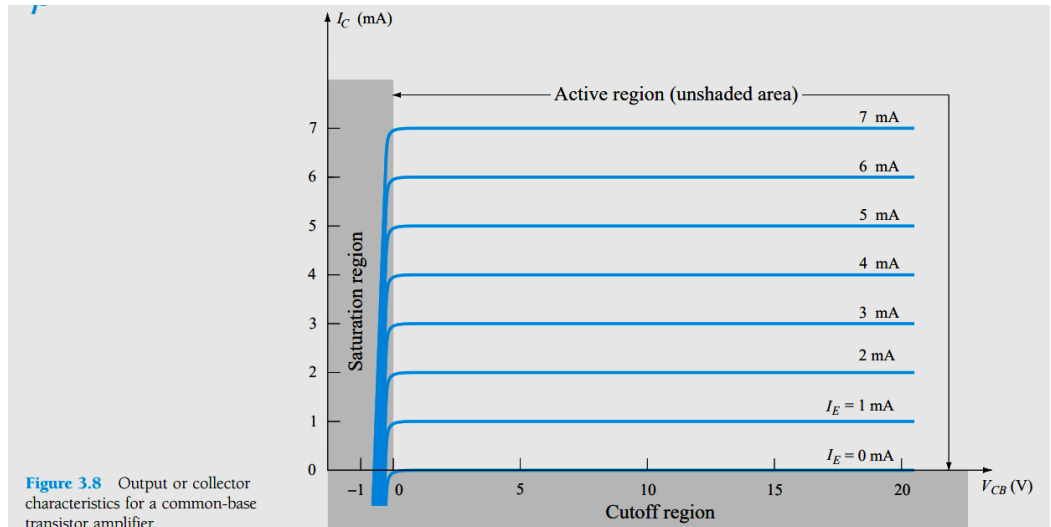
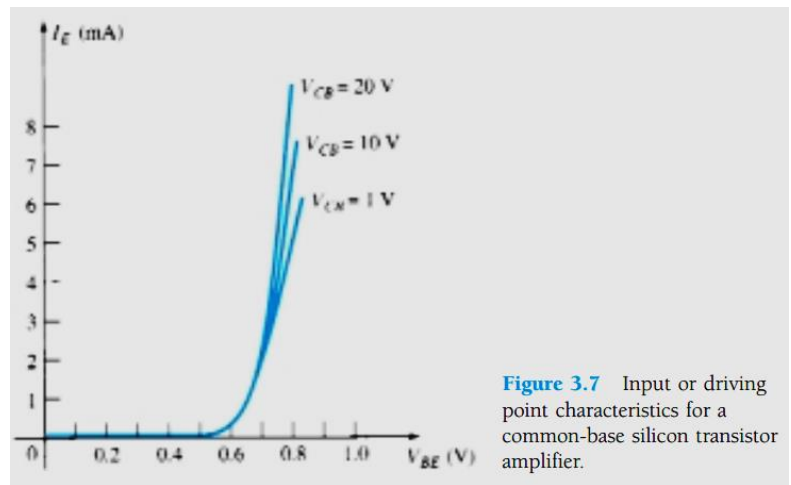
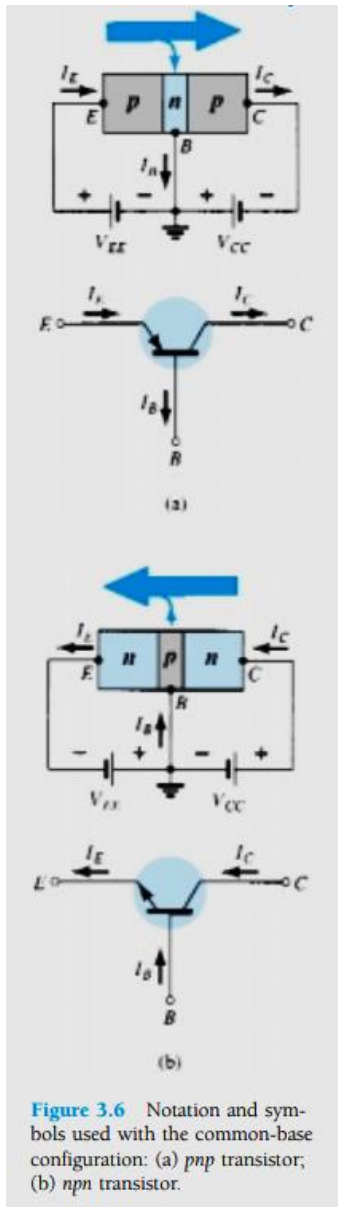
and find that the emitter current is the sum of the collector and base currents. The collector current, however, is comprised of two components—the majority and minority carriers as indicated in Fig. 3.5. The minority-current component is called the *leakage current* and is given the symbol I_{CO} (I_C current with emitter terminal Open). The collector current, therefore, is determined in total by Eq. (3.2).

$$I_C = I_{C_{\text{majority}}} + I_{CO_{\text{minority}}} \quad (3.2)$$

3.4 COMMON-BASE CONFIGURATION

The notation and symbols used in conjunction with the transistor in the majority of texts and manuals published today are indicated in Fig. 3.6 for the common-base configuration with *pnp* and *npn* transistors. The common-base terminology is derived from the fact that the base is common to both the input and output sides of the configuration. In addition, the base is usually the terminal closest to, or at, ground potential. Throughout this book all current directions will refer to conventional (hole) flow rather than electron flow. This choice was based primarily on the fact that the vast amount of literature available at educational and industrial institutions employs conventional flow and the arrows in all electronic symbols have a direction defined by this convention. Recall that the arrow in the diode symbol defined the direction of conduction for conventional current. For the transistor:

The arrow in the graphic symbol defines the direction of emitter current (conventional flow) through the device.



cutoff, and saturation regions. The active region is the region normally employed for linear (undistorted) amplifiers. In particular:

In the active region the collector-base junction is reverse-biased, while the base-emitter junction is forward-biased.

Note in Fig. 3.8 that as the emitter current increases above zero, the collector current increases to a magnitude essentially equal to that of the emitter current as determined by the basic transistor-current relations. Note also the almost negligible effect of V_{CB} on the collector current for the active region. The curves clearly indicate that *a first approximation to the relationship between I_E and I_C in the active region is given by*

$$I_C \cong I_E \quad (3.3)$$

As inferred by its name, the cutoff region is defined as that region where the collector current is 0 A, as revealed on Fig. 3.8. In addition:

In the cutoff region the collector-base and base-emitter junctions of a transistor are both reverse-biased.

The saturation region is defined as that region of the characteristics to the left of $V_{CB} = 0$ V. The horizontal scale in this region was expanded to clearly show the dramatic change in characteristics in this region. Note the exponential increase in collector current as the voltage V_{CB} increases toward 0 V.

In the saturation region the collector-base and base-emitter junctions are forward-biased.

3.5 TRANSISTOR AMPLIFYING ACTION

Now that the relationship between I_C and I_E has been established in Section 3.4, the basic amplifying action of the transistor can be introduced on a surface level using the network of Fig. 3.12. The dc biasing does not appear in the figure since our interest will be limited to the ac response. For the common-base configuration the ac input resistance determined by the characteristics of Fig. 3.7 is quite small and typically varies from 10 to 100 Ω . The output resistance as determined by the curves of Fig. 3.8 is quite high (the more horizontal the curves the higher the resistance) and typically varies from 50 k Ω to 1 M Ω (100 k Ω for the transistor of Fig. 3.12). The difference in resistance is due to the forward-biased junction at the input (base to emitter) and the reverse-biased junction at the output (base to collector). Using a common value of 20 Ω for the input resistance, we find that

$$I_i = \frac{V_i}{R_i} = \frac{200 \text{ mV}}{20 \Omega} = 10 \text{ mA}$$

If we assume for the moment that $\alpha_{ac} = 1$ ($I_C = I_e$),

$$I_L = I_i = 10 \text{ mA}$$

and

$$\begin{aligned} V_L &= I_L R \\ &= (10 \text{ mA})(5 \text{ k}\Omega) \\ &= 50 \text{ V} \end{aligned}$$

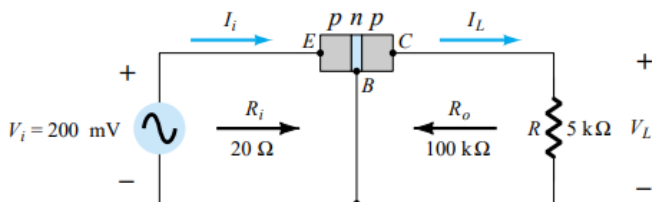


Figure 3.12 Basic voltage amplification action of the common-base configuration.

The voltage amplification is

$$A_v = \frac{V_L}{V_i} = \frac{50 \text{ V}}{200 \text{ mV}} = 250$$

Typical values of voltage amplification for the common-base configuration vary from 50 to 300. The current amplification (I_C/I_E) is always less than 1 for the common-base configuration. This latter characteristic should be obvious since $I_C = \alpha I_E$ and α is always less than 1.

The basic amplifying action was produced by transferring a current I from a low- to a high-resistance circuit. The combination of the two terms in italics results in the label *transistor*; that is,

transfer + resistor → transistor

3.6 COMMON-EMITTER CONFIGURATION

The most frequently encountered transistor configuration appears in Fig. 3.13 for the *pnp* and *npn* transistors. It is called the *common-emitter configuration* since the emitter is common or reference to both the input and output terminals (in this case common to both the base and collector terminals). Two sets of characteristics are again necessary to describe fully the behavior of the common-emitter configuration: one for the *input* or *base-emitter* circuit and one for the *output* or *collector-emitter* circuit. Both are shown in Fig. 3.14.

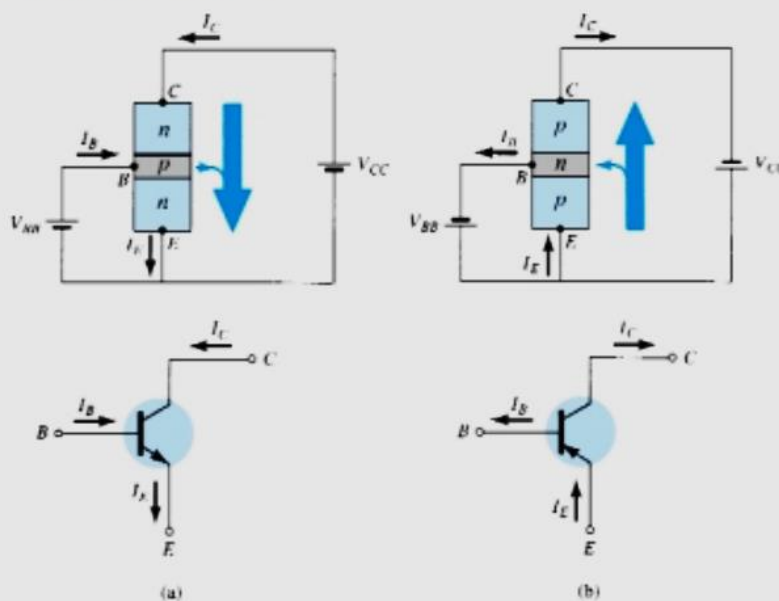


Figure 3.13 Notation and symbols used with the common-emitter configuration: (a) *nnp* transistor; (b) *pnp* transistor.

For the common-emitter configuration the output characteristics are a plot of the output current (I_C) versus output voltage (V_{CE}) for a range of values of input current (I_B). The input characteristics are a plot of the input current (I_B) versus the input voltage (V_{BE}) for a range of values of output voltage (V_{CE}).

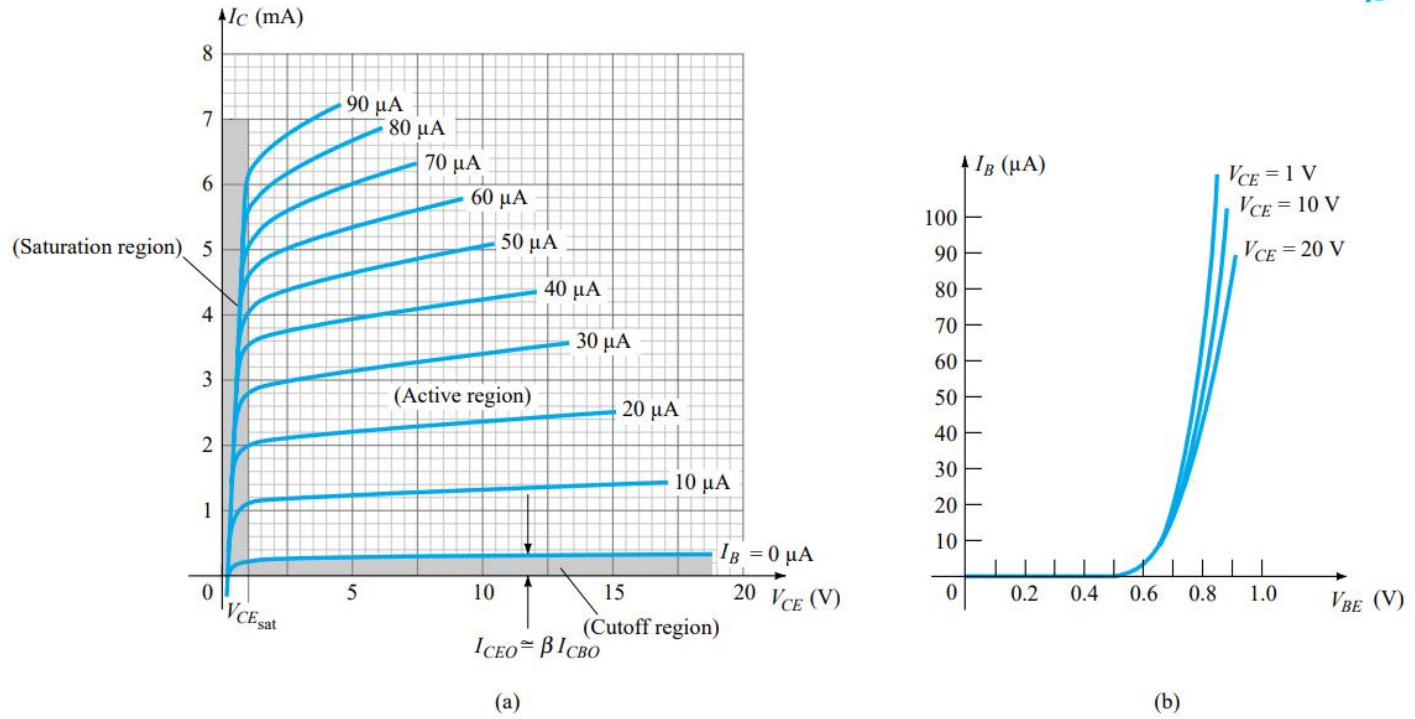


Figure 3.14 Characteristics of a silicon transistor in the common-emitter configuration: (a) collector characteristics; (b) base characteristics.

The active region for the common-emitter configuration is that portion of the upper-right quadrant that has the greatest linearity, that is, that region in which the curves for I_B are nearly straight and equally spaced. In Fig. 3.14a this region exists to the right of the vertical dashed line at V_{CEsat} and above the curve for I_B equal to zero. The region to the left of V_{CEsat} is called the saturation region.

In the active region of a common-emitter amplifier the collector-base junction is reverse-biased, while the base-emitter junction is forward-biased.

A relationship can be developed between β and α using the basic relationships introduced thus far. Using $\beta = I_C/I_B$ we have $I_B = I_C/\beta$, and from $\alpha = I_C/I_E$ we have $I_E = I_C/\alpha$. Substituting into

$$I_E = I_C + I_B$$

we have

$$\frac{I_C}{\alpha} = I_C + \frac{I_C}{\beta}$$

and dividing both sides of the equation by I_C will result in

$$\frac{1}{\alpha} = 1 + \frac{1}{\beta}$$

or

$$\beta = \alpha\beta + \alpha = (\beta + 1)\alpha$$

so that

$$\alpha = \frac{\beta}{\beta + 1} \quad (3.12a)$$

or

$$\beta = \frac{\alpha}{1 - \alpha} \quad (3.12b)$$

3.7 COMMON-COLLECTOR CONFIGURATION

The third and final transistor configuration is the *common-collector configuration*, shown in Fig. 3.20 with the proper current directions and voltage notation. The common-collector configuration is used primarily for impedance-matching purposes since it has a high input impedance and low output impedance, opposite to that of the common-base and common-emitter configurations.

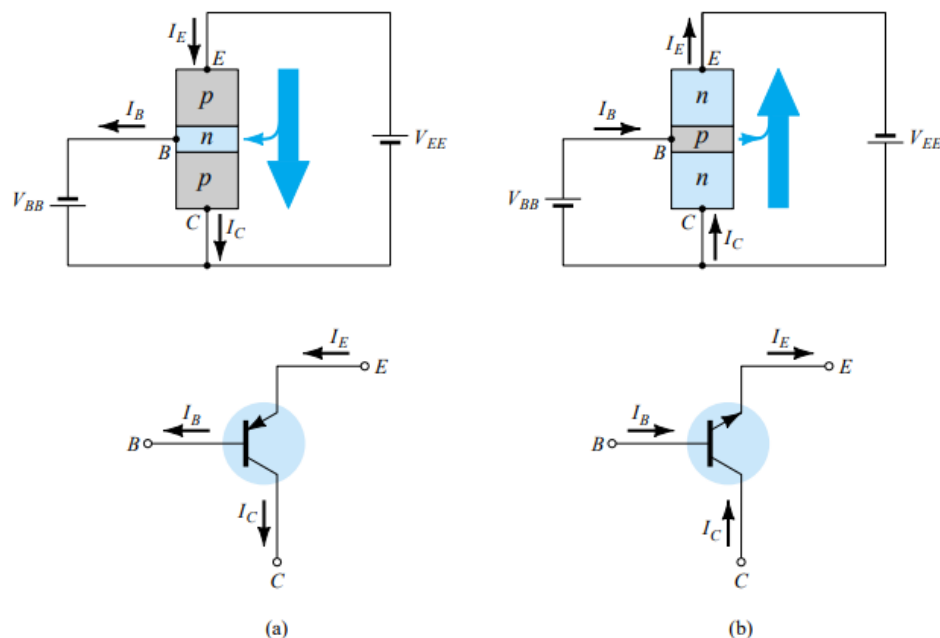


Figure 3.20 Notation and symbols used with the common-collector configuration: (a) pnp transistor; (b) npn transistor.

4.2 OPERATING POINT

The term *biasing* appearing in the title of this chapter is an all-inclusive term for the application of dc voltages to establish a fixed level of current and voltage. For transistor amplifiers the resulting dc current and voltage establish an *operating point* on the characteristics that define the region that will be employed for amplification of the applied signal. Since the operating point is a fixed point on the characteristics, it is also called the *quiescent point* (abbreviated *Q*-point). By definition, *quiescent* means quiet, still, inactive. Figure 4.1 shows a general output device characteristic with four operating points indicated. The biasing circuit can be designed to set the device operation at any of these points or others within the *active region*. The maximum ratings are indicated on the characteristics of Fig. 4.1 by a horizontal line for the maximum collector current $I_{C_{max}}$ and a vertical line at the maximum collector-to-emitter voltage $V_{CE_{max}}$. The maximum power constraint is defined by the curve $P_{C_{max}}$ in the same figure. At the lower end of the scales are the *cutoff region*, defined by $I_B \leq 0 \mu A$, and the *saturation region*, defined by $V_{CE} \leq V_{CE_{sat}}$.

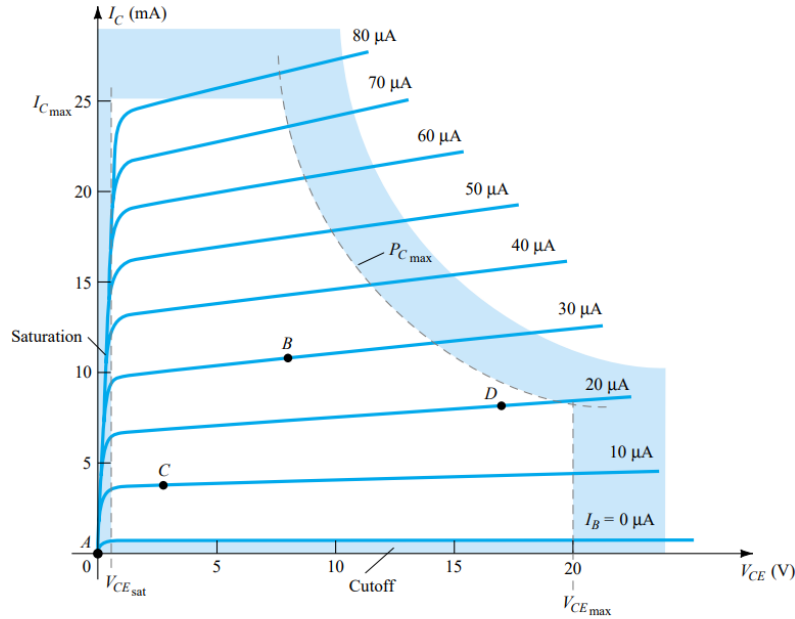


Figure 4.1 Various operating points within the limits of operation of a transistor.

Load-Line Analysis

Fig. 4.11a establishes an output equation that relates the variables I_C and V_{CE} in the following manner:

$$V_{CE} = V_{CC} - I_C R_C \quad (4.12)$$

The device characteristics of I_C versus V_{CE} are provided in Fig. 4.11b. We must now superimpose the straight line defined by Eq. (4.12) on the characteristics. The most direct method of plotting Eq. (4.12) on the output characteristics is to use the fact that a straight line is defined by two points. If we choose I_C to be 0 mA, we are specifying the horizontal axis as the line on which one point is located. By substituting $I_C = 0$ mA into Eq. (4.12), we find that

$$V_{CE} = V_{CC} - (0)R_C$$

and

$$V_{CE} = V_{CC} |_{I_C=0 \text{ mA}} \quad (4.13)$$

defining one point for the straight line as shown in Fig. 4.12.

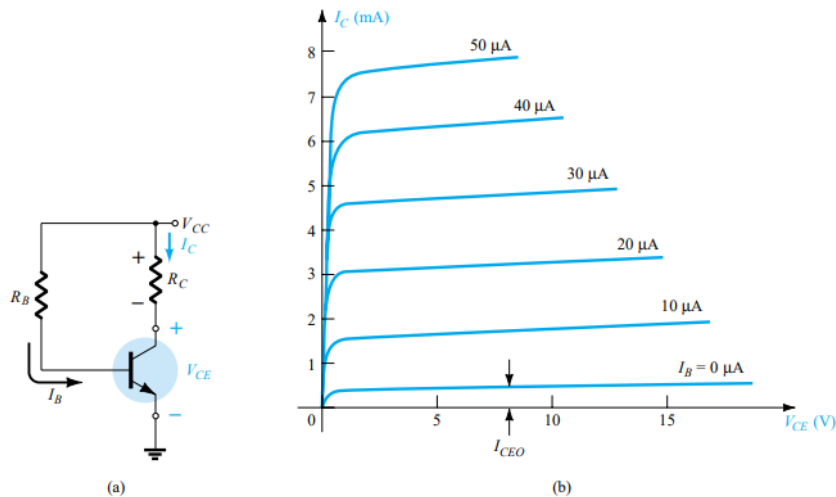


Figure 4.11 Load-line analysis: (a) the network; (b) the device characteristics.

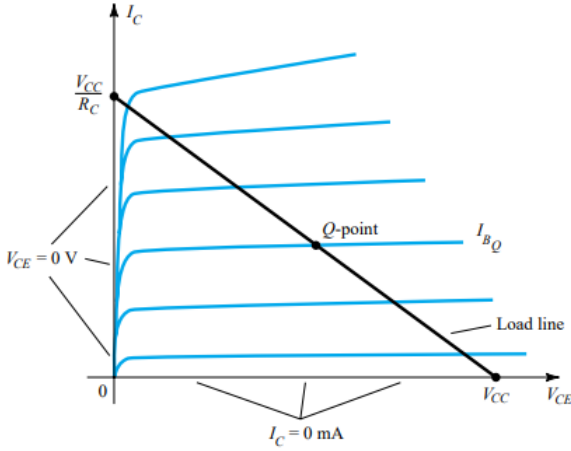


Figure 4.12 Fixed-bias load line.

If we now choose V_{CE} to be 0 V, which establishes the vertical axis as the line on which the second point will be defined, we find that I_C is determined by the following equation:

$$0 = V_{CC} - I_C R_C$$

and

$$I_C = \frac{V_{CC}}{R_C} \bigg|_{V_{ce} = 0 \text{ V}} \quad (4.14)$$

5.1 INTRODUCTION

The field-effect transistor (FET) is a three-terminal device used for a variety of applications that match, to a large extent, those of the BJT transistor described in Chapters 3 and 4. Although there are important differences between the two types of devices, there are also many similarities that will be pointed out in the sections to follow.

The primary difference between the two types of transistors is the fact that the BJT transistor is a *current-controlled* device as depicted in Fig. 5.1a, while the JFET transistor is a *voltage-controlled* device as shown in Fig. 5.1b. In other words, the current I_C in Fig. 5.1a is a direct function of the level of I_B . For the FET the current I will be a function of the voltage V_{GS} applied to the input circuit as shown in Fig. 5.1b. In each case the current of the output circuit is being controlled by a parameter of the input circuit—in one case a current level and in the other an applied voltage.

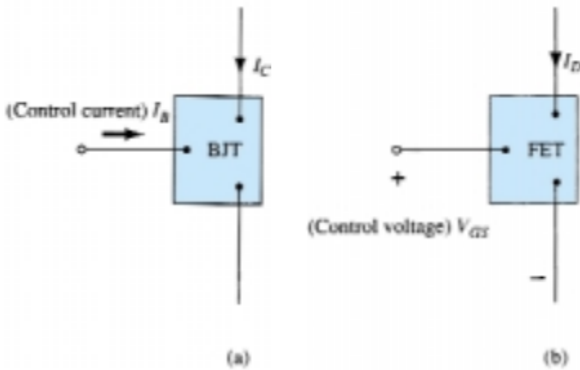


Figure 5.1 (a) Current-controlled and (b) voltage-controlled amplifiers.

Just as there are *nnp* and *pnnp* bipolar transistors, there are *n-channel* and *p-channel* field-effect transistors. However, it is important to keep in mind that the BJT transistor is a *bipolar* device—the prefix *bi-* revealing that the conduction level is a function of two charge carriers, electrons and holes. The FET is a *unipolar* device depending solely on either electron (*n-channel*) or hole (*p-channel*) conduction.

5.2 CONSTRUCTION AND CHARACTERISTICS OF JFETs

The basic construction of the n -channel JFET is shown in Fig. 5.2. Note that the major part of the structure is the n -type material that forms the channel between the embedded layers of p -type material. The top of the n -type channel is connected through an ohmic contact to a terminal referred to as the *drain* (D), while the lower end of the same material is connected through an ohmic contact to a terminal referred to as the *source* (S). The two p -type materials are connected together and to the *gate* (G) terminal. In essence, therefore, the drain and source are connected to the ends of the n -type channel and the gate to the two layers of p -type material. In the absence of any applied potentials the JFET has two p - n junctions under no-bias conditions. The result is a depletion region at each junction as shown in Fig. 5.2 that resembles the same region of a diode under no-bias conditions. Recall also that a depletion region is that region void of free carriers and therefore unable to support conduction through the region.

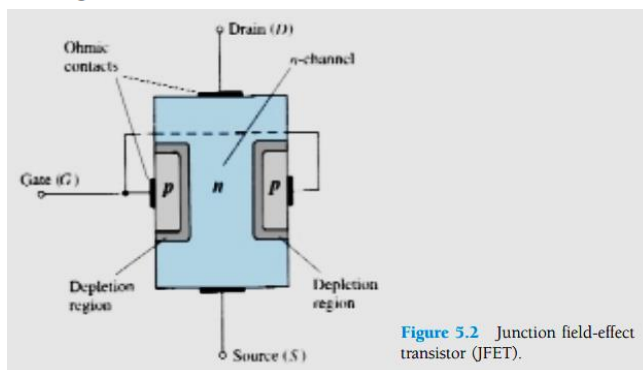


Figure 5.2 Junction field-effect transistor (JFET).

$V_{GS} = 0$ V, V_{DS} Some Positive Value

In Fig. 5.4, a positive voltage V_{DS} has been applied across the channel and the gate has been connected directly to the source to establish the condition $V_{GS} = 0$ V. The result is a gate and source terminal at the same potential and a depletion region in the low end of each p -material similar to the distribution of the no-bias conditions of Fig. 5.2. The instant the voltage $V_{DD} (= V_{DS})$ is applied, the electrons will be drawn to the drain terminal, establishing the conventional current I_D with the defined direction of Fig. 5.4. The path of charge flow clearly reveals that the drain and source currents are equivalent ($I_D = I_S$). Under the conditions appearing in Fig. 5.4, the flow of charge is relatively uninhibited and limited solely by the resistance of the n -channel between drain and source.

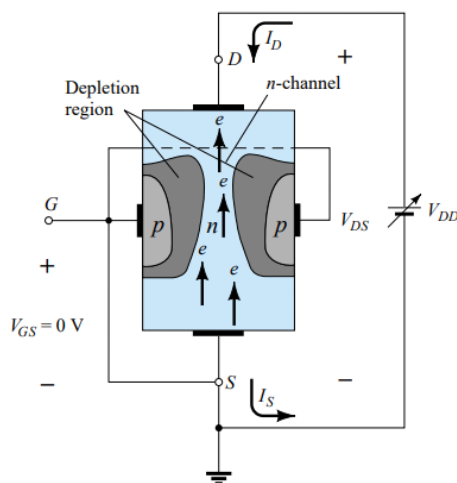


Figure 5.4 JFET in the $V_{GS} = 0$ V and $V_{DS} > 0$ V.

As the voltage V_{DS} is increased from 0 to a few volts, the current will increase as determined by Ohm's law and the plot of I_D versus V_{DS} will appear as shown in Fig. 5.6. The relative straightness of the plot reveals that for the region of low values of V_{DS} , the resistance is essentially constant. As V_{DS} increases and approaches a level referred to as V_P in Fig. 5.6, the depletion regions of Fig. 5.4 will widen, causing a noticeable reduction in the channel width. The reduced path of conduction causes the resistance to increase and the curve in the graph of Fig. 5.6 to occur. The more horizontal the curve, the higher the resistance, suggesting that the resistance is approaching "infinite" ohms in the horizontal region. If V_{DS} is increased to a level where it appears that the two depletion regions would "touch" as shown in Fig. 5.7, a condition referred to as *pinch-off* will result. The level of V_{DS} that establishes this condition is referred to as the *pinch-off voltage* and is denoted by V_P as shown in Fig. 5.6. In actuality, the term *pinch-off* is a misnomer in that it suggests the current I_D is pinched off and drops to 0 A. As shown in Fig. 5.6, however, this is hardly the case— I_D maintains a saturation level defined as I_{DSS} in Fig. 5.6. In reality a very small channel still exists, with a current of very high density. The fact that I_D does not drop off at pinch-off and maintains the saturation level indicated in Fig. 5.6 is verified by the following fact: The absence of a drain current would remove the possibility of different potential levels through the n -channel material to establish the varying levels of reverse bias along the p - n junction. The result would be a loss of the depletion region distribution that caused pinch-off in the first place.

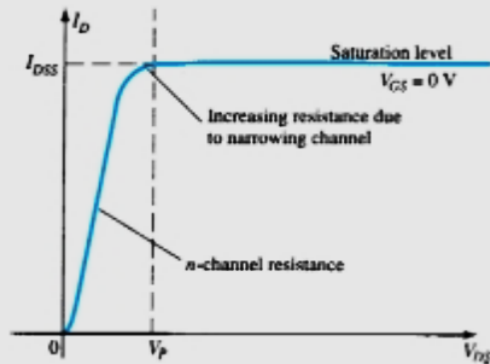


Figure 5.6 I_D versus V_{DS} for $V_{GS} = 0$ V.

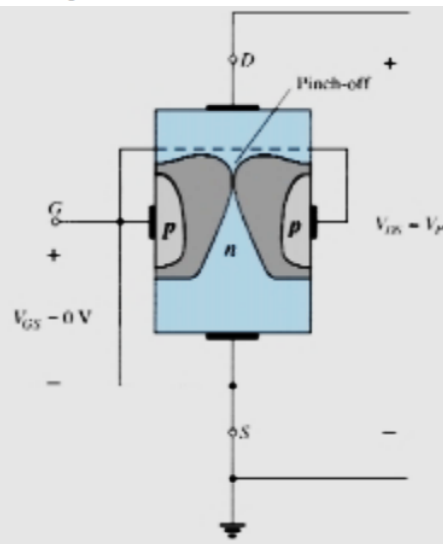


Figure 5.7 Pinch-off ($V_{GS} = 0$ V, $V_{DS} = V_P$).

5.7 DEPLETION-TYPE MOSFET

As noted in the chapter introduction, there are two types of FETs: JFETs and MOSFETs. MOSFETs are further broken down into *depletion type* and *enhancement type*. The terms *depletion* and *enhancement* define their basic mode of operation, while the label MOSFET stands for *metal-oxide-semiconductor-field-effect transistor*. Since there are differences in the characteristics and operation of each type of MOSFET, they are covered in separate sections. In this section we examine the depletion-type MOSFET, which happens to have characteristics similar to those of a JFET between cutoff and saturation at I_{DSS} but then has the added feature of characteristics that extend into the region of opposite polarity for V_{GS} .

Basic Construction

The basic construction of the n -channel depletion-type MOSFET is provided in Fig. 5.23. A slab of p -type material is formed from a silicon base and is referred to as the *substrate*. It is the foundation upon which the device will be constructed. In some cases the substrate is internally connected to the source terminal. However, many discrete devices provide an additional terminal labeled SS , resulting in a four-terminal device, such as that appearing in Fig. 5.23. The source and drain terminals are connected through metallic contacts to n -doped regions linked by an n -channel as shown in the figure. The gate is also connected to a metal contact surface but remains insulated from the n -channel by a very thin silicon dioxide (SiO_2) layer. SiO_2 is a particular type of insulator referred to as a *dielectric* that sets up opposing (as revealed by

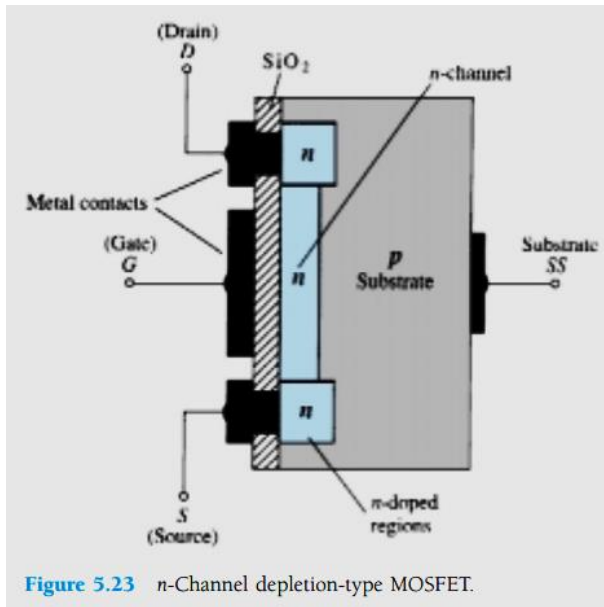


Figure 5.23 n -Channel depletion-type MOSFET.

the prefix *di-*) electric fields within the dielectric when exposed to an externally applied field. The fact that the SiO_2 layer is an insulating layer reveals the following fact:

There is no direct electrical connection between the gate terminal and the channel of a MOSFET.

In addition:

It is the insulating layer of SiO_2 in the MOSFET construction that accounts for the very desirable high input impedance of the device.

Basic Operation and Characteristics

In Fig. 5.24 the gate-to-source voltage is set to zero volts by the direct connection from one terminal to the other, and a voltage V_{DS} is applied across the drain-to-source terminals. The result is an attraction for the positive potential at the drain by the free electrons of the n -channel and a current similar to that established through the channel of the JFET. In fact, the resulting current with $V_{GS} = 0$ V continues to be labeled I_{DSS} , as shown in Fig. 5.25.

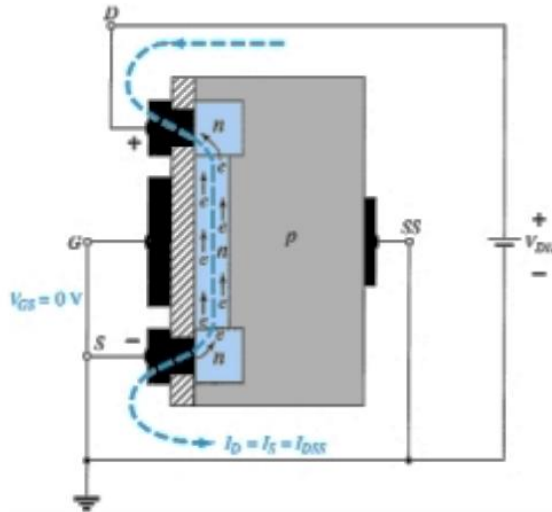


Figure 5.24 n -Channel depletion-type MOSFET with $V_{GS} = 0$ V and an applied voltage V_{DD} .

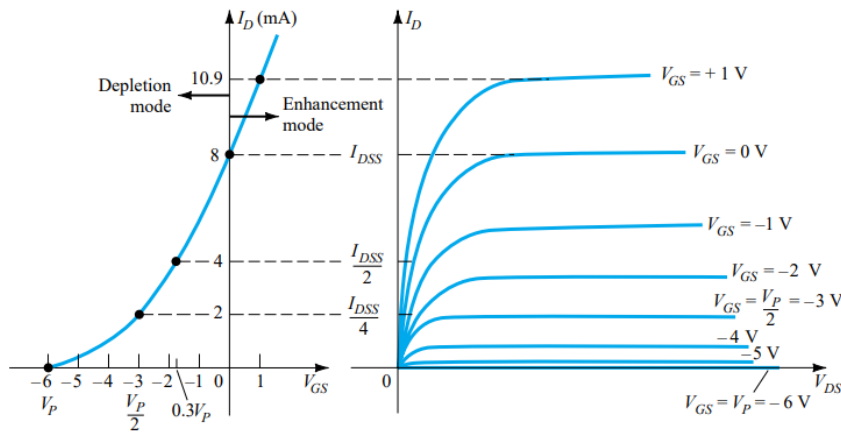


Figure 5.25 Drain and transfer characteristics for an n -channel depletion-type MOSFET.

In Fig. 5.26, V_{GS} has been set at a negative voltage such as -1 V. The negative potential at the gate will tend to pressure electrons toward the p -type substrate (like charges repel) and attract holes from the p -type substrate (opposite charges attract) as shown in Fig. 5.26. Depending on the magnitude of the negative bias established by V_{GS} , a level of recombination between electrons and holes will occur that will reduce the number of free electrons in the n -channel available for conduction. The more negative the bias, the higher the rate of recombination. The resulting level of drain current is therefore reduced with increasing negative bias for V_{GS} as shown in Fig. 5.25 for $V_{GS} = -1$ V, -2 V, and so on, to the pinch-off level of -6 V. The resulting levels of drain current and the plotting of the transfer curve proceeds exactly as described for the JFET.

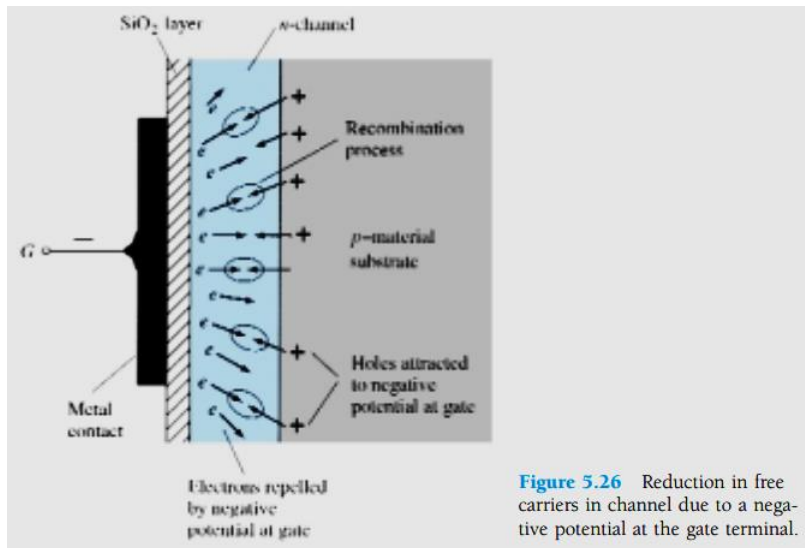


Figure 5.26 Reduction in free carriers in channel due to a negative potential at the gate terminal.

5.8 ENHANCEMENT-TYPE MOSFET

Although there are some similarities in construction and mode of operation between depletion-type and enhancement-type MOSFETs, the characteristics of the enhancement-type MOSFET are quite different from anything obtained thus far. The transfer curve is not defined by Shockley's equation, and the drain current is now cut off until the gate-to-source voltage reaches a specific magnitude. In particular, current control in an n -channel device is now effected by a positive gate-to-source voltage rather than the range of negative voltages encountered for n -channel JFETs and n -channel depletion-type MOSFETs.

Basic Construction

The basic construction of the n -channel enhancement-type MOSFET is provided in Fig. 5.31. A slab of p -type material is formed from a silicon base and is again referred to as the substrate. As with the depletion-type MOSFET, the substrate is sometimes internally connected to the source terminal, while in other cases a fourth lead is made available for external control of its potential level. The source and drain terminals are again connected through metallic contacts to n -doped regions, but note in Fig. 5.31 the absence of a channel between the two n -doped regions. This is the primary difference between the construction of depletion-type and enhancement-type MOSFETs—the absence of a channel as a constructed component of the device. The SiO_2 layer is still present to isolate the gate metallic platform from the region between the drain and source, but now it is simply separated from a section of the p -type material. In summary, therefore, the construction of an enhancement-type MOSFET is quite similar to that of the depletion-type MOSFET, except for the absence of a channel between the drain and source terminals.

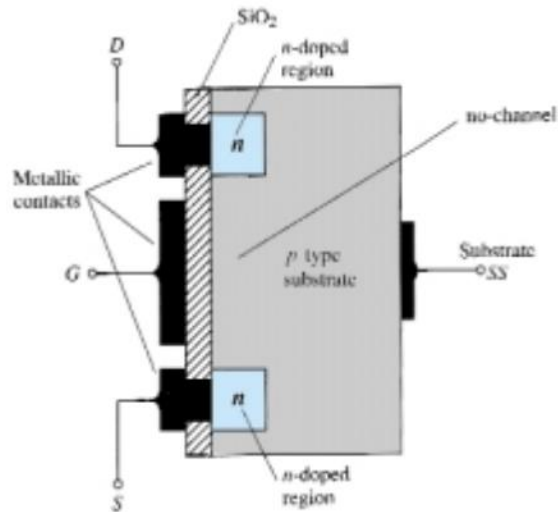


Figure 5.31 n-Channel enhancement-type MOSFET.

Basic Operation and Characteristics

In Fig. 5.32 both V_{DS} and V_{GS} have been set at some positive voltage greater than 0 V, establishing the drain and gate at a positive potential with respect to the source. The positive potential at the gate will pressure the holes (since like charges repel) in the p -substrate along the edge of the SiO_2 layer to leave the area and enter deeper regions of the p -substrate, as shown in the figure. The result is a depletion region near the SiO_2 insulating layer void of holes. However, the electrons in the p -substrate (the minority carriers of the material) will be attracted to the positive gate and accumulate in the region near the surface of the SiO_2 layer. The SiO_2 layer and its insulating qualities will prevent the negative carriers from being absorbed at the gate terminal. As V_{GS} increases in magnitude, the concentration of electrons near the SiO_2 surface increases until eventually the induced n -type region can support a measurable flow between drain and source. The level of V_{GS} that results in the significant increase in drain current is called the *threshold voltage* and is given the symbol V_T . On specification sheets it is referred to as $V_{GS(\text{Th})}$, although V_T is less unwieldy and will be used in the analysis to follow. Since the channel is nonexistent with $V_{GS} = 0$ V and “enhanced” by the application of a positive gate-to-source voltage, this type of MOSFET is called an *enhancement-type MOSFET*. Both depletion- and enhancement-type MOSFETs have enhancement-type regions, but the label was applied to the latter since it is its only mode of operation.

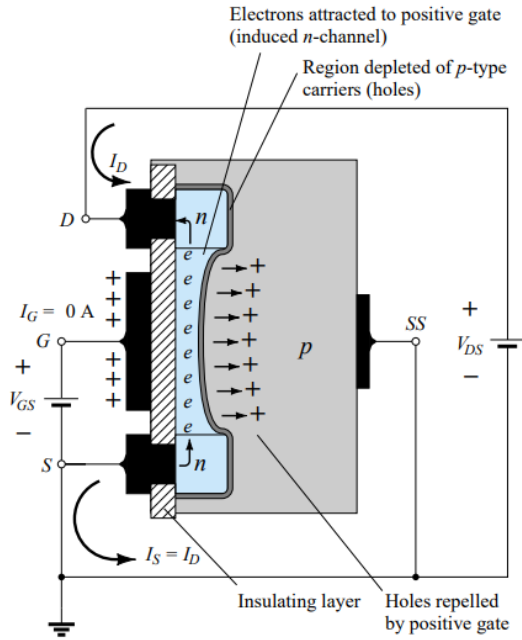


Figure 5.32 Channel formation in the n -channel enhancement-type MOSFET.

As V_{GS} is increased beyond the threshold level, the density of free carriers in the induced channel will increase, resulting in an increased level of drain current. However, if we hold V_{GS} constant and increase the level of V_{DS} , the drain current will eventually reach a saturation level as occurred for the JFET and depletion-type MOSFET. The leveling off of I_D is due to a pinching-off process depicted by the narrower channel at the drain end of the induced channel as shown in Fig. 5.33. Applying Kirchhoff's voltage law to the terminal voltages of the MOSFET of Fig. 5.33, we find that

$$V_{DG} = V_{DS} - V_{GS} \quad (5.11)$$